



Department of Electronic & Telecommunication Engineering
University of Moratuwa
Sri Lanka.

Project Report

Smart Power Quality Analyzer

EN2160 - Electronic Design Realization

Group Name: ElectroSquad

Anthony C.S.B.	: 230045X
Chandrakumara H.A.D.C.	: 230100M
Hansindu M.M.A.D.	: 230229P
Perera H.A.J.I.	: 230477X
Weerasinghe J.A.H.R.	: 230697V

Date: 20-07-2026

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Smart Power Quality Analyzer - Project Report

1 Problem Description, Justification and Benefits of Our Solution

1.1 Problem Description

Modern electrical systems frequently experience power-quality problems such as voltage sags, voltage swells, harmonic distortion, poor power factor, and frequency variations. These problems can reduce equipment performance, increase energy losses, shorten equipment lifetime, and cause unexpected failures.

Commercial power-quality analyzers are often expensive and may not be affordable for small industries, laboratories, educational institutions, and domestic users. Therefore, there is a need for a low-cost and user-friendly system capable of monitoring important electrical parameters in real time.

1.2 Our Solution

The proposed solution is a **Smart Power Quality Analyzer** that measures, processes, displays, stores, and remotely presents important power-quality parameters.

The system is designed to monitor:

- RMS voltage;
- RMS current;
- frequency;
- active power;
- reactive power;
- apparent power;
- power factor;
- voltage THD;
- current THD;
- voltage harmonics from H1 to H25;
- current harmonics from H1 to H25; and
- voltage and current waveforms.

1.3 Justification

The proposed system is justified by the need for an affordable, accessible, and portable power-quality monitoring solution. In the Sri Lankan context, rising energy costs, limited access to expensive test equipment, and the increasing use of nonlinear electrical loads create a clear need for such a device.

The system combines local monitoring through a TFT display with remote monitoring through a web dashboard. This makes the device suitable for educational, industrial, laboratory, and domestic applications.

1.4 Benefits of Our Solution

- Early detection of abnormal electrical conditions.
- Improved protection of electrical and electronic equipment.
- Better understanding of energy consumption.
- Real-time local and remote monitoring.
- Lower cost than commercial analyzers.
- Availability of logged data for later analysis.
- Easy access to completed sessions through CSV files.
- Suitability for educational and small-scale industrial use.

2 Technical Feasibility

2.1 Hardware Feasibility

The system uses commonly available microcontrollers and electronic components. The STM32 controller is used for signal acquisition and electrical calculations, while the ESP32-S3 is used for display control, Wi-Fi communication, Firebase integration, and data logging. For better accuracy, a 16-bit external ADC is used rather than the internal STM ADC.

The main hardware components include:

- STM32 microcontroller
- ESP32-S3 microcontroller
- ADS8588 ADC
- voltage-sensing circuit
- current-sensing circuit
- TFT colour display
- FT6336 touch controller
- rotary encoder and control buttons
- power-supply circuit
- printed circuit board
- protection components
- product enclosure.

2.2 Software Feasibility

The firmware was developed using standard embedded system development tools, while the web dashboard was implemented using React, TypeScript and Vite. Firebase Realtime Database was used to store and retrieve live measurements and logged data in CSV format. The use of Firebase also enabled the deployment of a hosted web platform in which each analyzer can be identified through a unique device ID and accessed only through authorized email accounts assigned to that device.

2.3 Operational Feasibility

The system is designed to be simple to operate. The user can view live measurements on the TFT screen, navigate through waveform and harmonic screens, start and stop logging, synchronize completed sessions, and view data remotely through the dashboard.

2.4 Economic Feasibility

The proposed analyzer is expected to cost significantly less than commercial power quality analyzers. The design uses affordable and widely available components while still providing the main features required for real time monitoring and data logging.

3 Technical Specification

Table 1: Technical specifications of the proposed analyzer

Parameter	Specification
System type	Single-phase power-quality analyzer
Nominal voltage	230 V AC
Nominal frequency	50 Hz
Voltage measurement range	0-300 V
Current measurement range	0-50 A
Harmonic measurement range	H1 to H25
UART baud rate	921600 bits per second
Local display	TFT colour display
Wireless communication	Wi-Fi
Cloud platform	Firebase Realtime Database
Local storage	ESP32-S3 flash memory
Data-export format	CSV

3.1 Measured Parameters

Table 2: Measured and calculated electrical parameters

Parameter	Unit	Description
RMS voltage	V	Effective value of the measured voltage
RMS current	A	Effective value of the measured current
Frequency	Hz	Fundamental frequency of the supply
Active power	W	Real power consumed by the load
Reactive power	var	Reactive component of power
Apparent power	VA	Product of RMS voltage and RMS current
Power factor	–	Ratio of active power to apparent power
Voltage THD	%	Total harmonic distortion of voltage
Current THD	%	Total harmonic distortion of current
Voltage harmonics	%	Harmonic magnitudes from H1 to H25
Current harmonics	%	Harmonic magnitudes from H1 to H25

4 Product Architecture

4.1 Overall Architecture

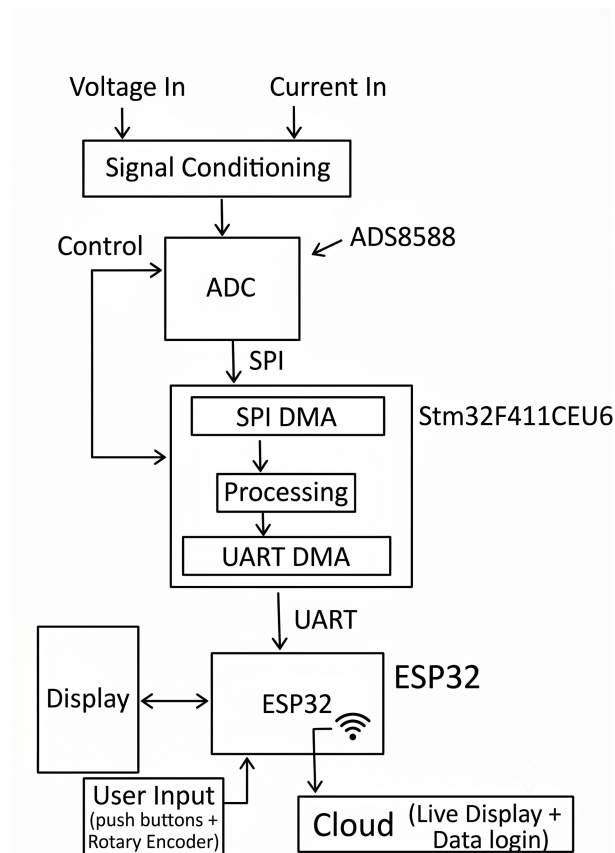


Figure 1: Overall product architecture

4.2 Voltage-Sensing Unit

The voltage-sensing unit uses a precision resistor divider to step down the mains voltage to a 250 mV differential signal. To ensure safety, the PCB segregates the circuitry into an isolated ground plane (referenced to mains neutral) and a main system ground plane. An AMC1301 isolation amplifier bridges these planes, providing galvanic isolation while transmitting the 250 mV signal. An OPA197 operational amplifier then acts as a differential amplifier to apply a precise gain, optimizing the signal for the maximum dynamic range of the ADS8588 ADC. Finally, a symmetric low-pass anti-aliasing filter removes high-frequency noise prior to digitization.

4.3 Current-Sensing Unit

The current-sensing unit measures load current using an SCT-013-050 current transformer. Because the sensor inherently provides galvanic isolation, its output is referenced directly to the main system ground plane of the PCB. This signal is routed to an OPA197 operational amplifier configured as a differential amplifier, which applies the specific gain required to utilize the optimal maximum range of the ADS8588 ADC. The output is then passed through a symmetric low-pass anti-aliasing filter directly to the ADC to ensure clean and accurate signal conversion.

4.4 STM32 Processing Unit

The STM32 microcontroller controls the ADC to sample voltage and current signals at a sampling frequency of 10 kHz. The digitized samples are acquired via the SPI interface. Every 102.4 ms, after collecting 1024 samples of both voltage and current, the microcontroller processes the data to compute various electrical parameters, including voltage RMS, current RMS, active power, apparent power, reactive power, power factor, frequency, voltage and current crest factors, swell factor, voltage and current total harmonic distortion (THD), and voltage and current harmonic magnitudes up to the 25th order. To improve computational efficiency and achieve real-time performance, the CMSIS-DSP library was utilized for the signal processing and mathematical computations. Upon completion of the calculations, the resulting measurement data is packaged and transmitted to the ESP32 microcontroller via UART at a baud rate of 921600 bps.

4.5 ESP32-S3 Unit

The ESP32-S3 receives processed data from the STM32 through UART. It validates the packet header, length, sequence number, and checksum before displaying or publishing the data.

- TFT display;
- touch input;
- rotary encoder and buttons;
- Wi-Fi connection;
- Firebase communication;
- flash-memory logging; and
- synchronization of completed sessions.

4.6 UART Communication

Table 3: UART connection between STM32 and ESP32-S3

STM32 Pin	ESP32-S3 Pin	Function
PA11	RX	STM32 transmission to ESP32-S3
PA12	TX	ESP32-S3 transmission to STM32
GND	GND	Common ground
3.3 V	3.3 V	Supply where applicable

4.7 Data Flow

1. Voltage and current signals are sensed.
2. The STM32 samples and processes the signals.
3. Calculated values, waveforms, and harmonics are placed in a UART packet.
4. The ESP32-S3 receives and validates the packet.
5. Valid data are displayed on the TFT screen.
6. Selected live values are uploaded to Firebase.
7. The web dashboard retrieves and displays the latest measurements.
8. Logging data are stored and later exported as CSV files.

5 Initial and Final Sketches of Product Enclosure

5.1 Initial Sketches

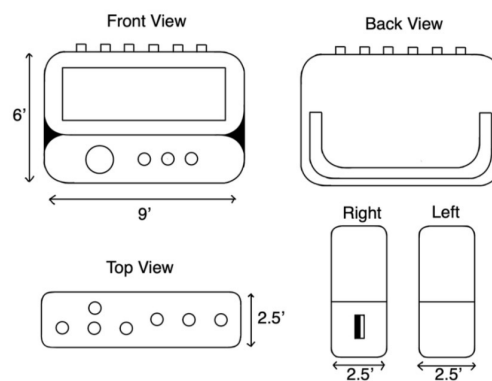
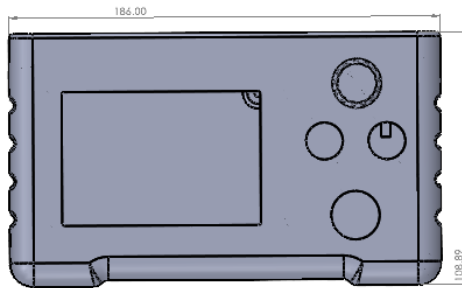
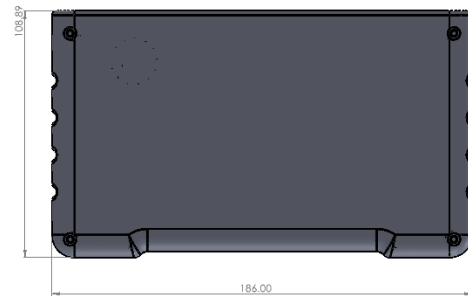


Figure 2: Initial Enclosure Sketches

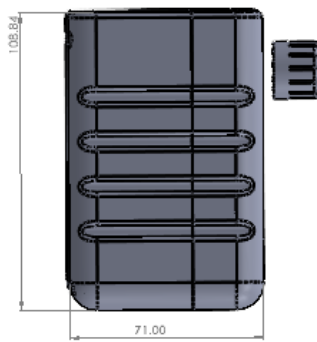
5.2 Final Design



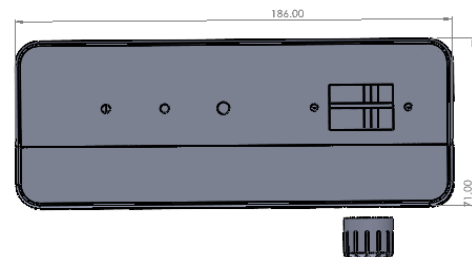
(a) Front view



(b) Back view



(c) Side view



(d) Top view

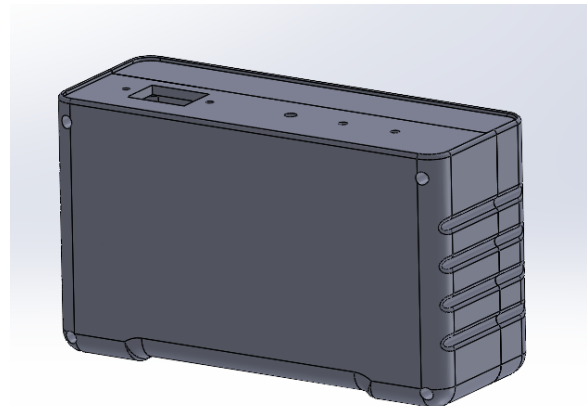
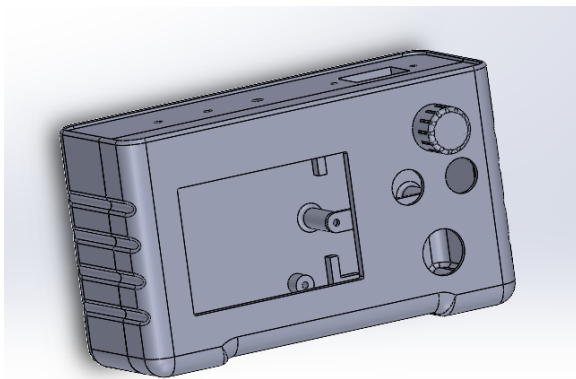


Figure 4: Enclosure from the front and back

6 Schematic and PCB Design

6.1 Schematic Design

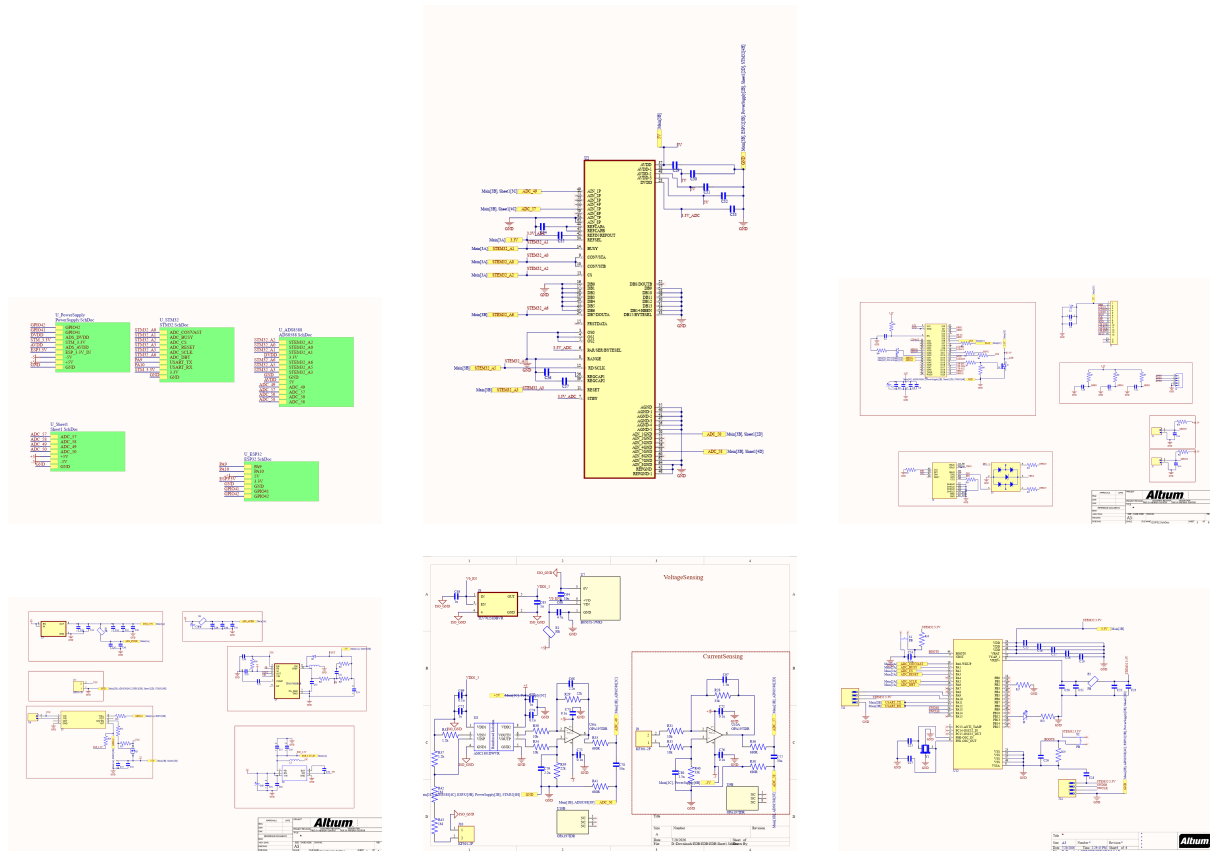
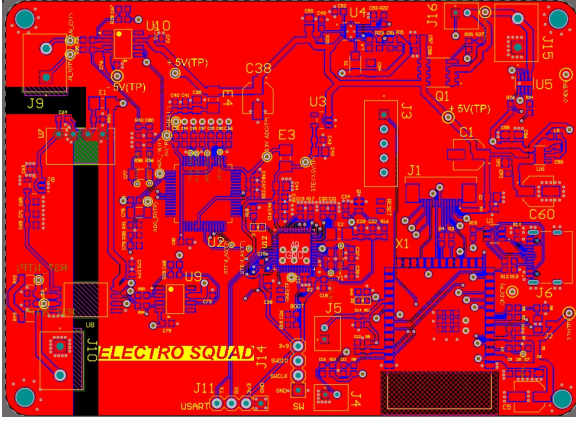
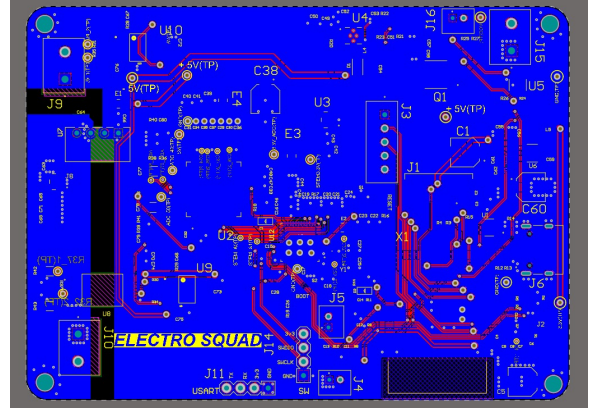


Figure 5: Complete schematic sheets of the Smart Power Quality Analyzer

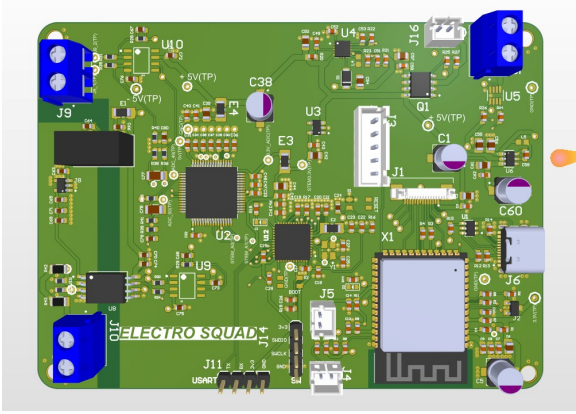
6.2 PCB Design



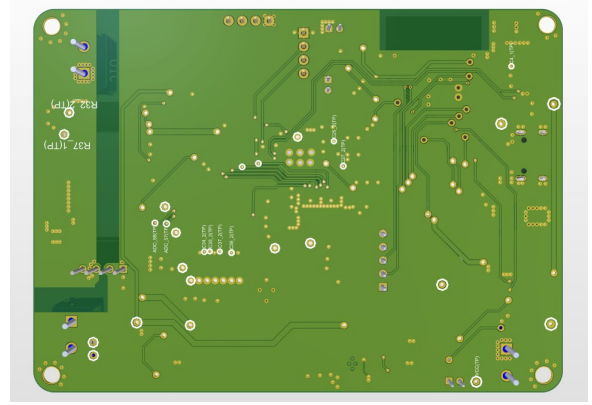
(a) Two-dimensional top view



(b) Two-dimensional bottom view



(c) Three-dimensional top view



(d) Three-dimensional bottom view

Figure 6: Two-dimensional and three-dimensional views of the PCB design

The two-dimensional views show the component placement, copper routing, ground planes, vias, and board outline on the top and bottom layers. The three-dimensional views were used to inspect component orientation, mechanical clearance, connector positions, and the overall physical arrangement of the assembled PCB.

6.3 PCB Design Features and Design Justification

1. Functional Subsystem Zoning

Components are grouped strategically by function—including the power supply, ESP32, STM32, ADC, and analog acquisition stages—to minimize routing complexity, shorten interconnects, and optimize structural layout.

2. Mixed-Signal Isolation

Analog acquisition and ADC stages are physically isolated from high-speed digital domains (ESP32, STM32, and LCD interfaces) to mitigate electromagnetic interference (EMI) and cross-talk, preserving measurement accuracy.

3. Decoupled Power Domain Distribution

Independent, regulated power networks service distinct subsystems: a buck converter

powers the ESP32, an LDO drives the STM32, and ferrite bead filtering isolates the analog ADC rail to ensure stable, low-noise supply lines.

4. **Minimized Switching Loop Geometry**

Buck converter topologies feature tight, clustered component placement to shrink high-current switching loops, reducing radiated switching noise, EMI, and parasitic power losses.

5. **Short-Path Analog Topology**

The ADS8588 ADC is positioned immediately adjacent to the signal conditioning circuitry to keep analog traces as short as possible, preventing noise pickup prior to digitization.

6. **Localized Decoupling Network**

Decoupling capacitors are placed at the immediate power pins of each integrated circuit to enhance power integrity by dampening transient voltage fluctuations.

7. **Low-Impedance Return Paths and Via Stitching**

Dual-layer ground polygon pours integrated with dense via stitching establish low-impedance return paths, optimizing current flow, lowering EMI, and stabilizing signal integrity.

8. **Antenna Keep-Out Constraints**

A dedicated, copper-free keep-out zone is maintained beneath the ESP32 antenna structure in compliance with vendor specifications to prevent RF detuning and maximize wireless performance.

9. **Low-Loss Power Trace Geometry**

High-current paths utilize widened trace geometries to diminish parasitic voltage drops, minimize internal power dissipation, and improve overall thermal efficiency.

10. **High-Speed Critical Signal Routing**

Critical high-speed lines, such as the STM32-to-ADS8588 SPI interface, are routed via short, direct trajectories over unbroken, continuous ground reference planes to preserve signal integrity.

11. **Aggressive Aggressor Isolation**

Known noise aggressors—such as switching regulators, USB interfaces, Wi-Fi modules, and LCD backlight circuitry—are placed away from sensitive analog measurement zones to block radiated and conducted interference.

12. **Diagnostic Test Point Integration**

Dedicated test points populate critical power rails, communication buses, and analog nodes to streamline voltage verification, signal diagnostics, and system debugging.

13. **Peripheral Accessibility and Ergonomics**

Human-machine interface (HMI) components, including display headers, USB ports, rotary encoders, and tactical switches, are aligned along the PCB perimeter for streamlined assembly and user interaction.

14. **Thermal Dissipation Management**

Thermal-generating elements like switching regulators are positioned near the board edge alongside expansive copper planes to accelerate heat dissipation away from temperature-sensitive analog components.

15. **Electromagnetic Compatibility (EMC) Optimization**

Strict enforcement of tight loop areas, continuous reference planes, shortened return paths, and explicit mixed-signal segregation ensures robust EMC performance and reliable hardware operation.

7 UI Design

7.1 TFT User Interface

Table 4: TFT screen functions

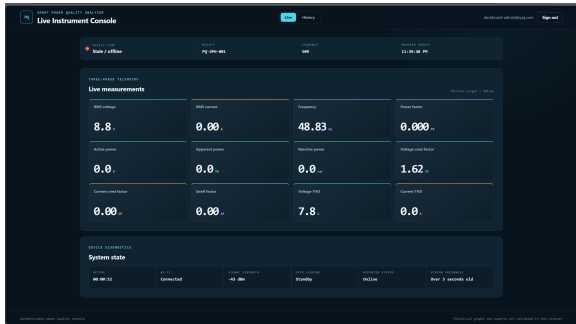
Screen		Function
Main screen	measurement	Displays important scalar measurements
Voltage screen	waveform	Displays the sampled voltage waveform
Current screen	waveform	Displays the sampled current waveform
Voltage screen	harmonics	Displays voltage harmonics from H1 to H25
Current screen	harmonics	Displays current harmonics from H1 to H25
Logging screen		Provides Start Log, Stop Log, and Push Sync controls
Wi-Fi screen		Displays connection and configuration information
Settings screen		Provides device settings and configuration options

7.2 Physical Controls

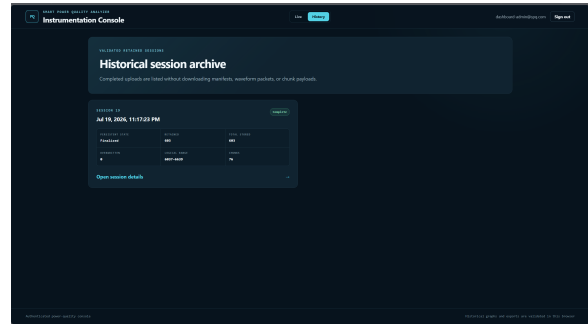
The physical controls include a rotary encoder, push buttons, and touch input. These controls allow the user to navigate between screens, start and stop logging, and synchronize stored data.

7.3 Web Dashboard

- Voltage.
- Current.
- Frequency.
- Power factor.
- THD.
- Active power.
- Device connection status.
- Warning and normal states.
- Trend graphs.
- Logging status.
- Completed sessions.
- CSV-download options.



(a) Live update web dashboard



(b) Log page in Website

Figure 7: Web Interface

7.4 Operating Procedure

1. Apply power to the analyzer.
2. Wait for the STM32 and ESP32-S3 to initialize.
3. Verify that live measurements are displayed.
4. Navigate through waveform, harmonic, and measurement screens.
5. Open the logging screen when data logging is required.
6. Press **Start Log**.
7. Press **Stop Log** when the required period is complete.
8. Press **Push Sync** to synchronize the session.
9. Access the web dashboard and download the CSV file.

8 Sales and Maintenance

8.1 Target Customers

- Educational laboratories.
- Small and medium-scale industries.
- Electrical technicians.
- Maintenance engineers.
- Energy auditors.
- Research laboratories.
- Domestic users.
- Renewable-energy system operators.

8.2 Sales Strategy

The product can be sold directly to educational institutions, laboratories, small industries, and technicians. Online platforms and an official product website may also be used for sales.

8.3 Packaging

- Product name.
- Product image.
- Technical specifications.
- Serial number.
- Manufacturing date.
- Safety instructions.
- QR code.
- Warranty details.
- Contact information.

8.4 Maintenance

The device should be periodically calibrated to maintain measurement accuracy. Firmware updates may be provided when improvements are made. Replaceable parts should be used wherever possible to simplify maintenance.

8.5 Repair and Warranty

A suitable warranty period should be provided. During the warranty period, manufacturing defects and technical faults should be repaired or the unit should be replaced when necessary.

9 Volume Production and Product Costing

9.1 Considerations for Volume Production

When the product enters volume production, the following factors should be considered:

- Bulk purchasing of electronic components to reduce material cost.
- Automated PCB assembly and testing to improve production efficiency.
- Standardized manufacturing processes to ensure consistent product quality.
- Design for Manufacturability (DFM) to simplify assembly and reduce production time.
- Quality control and calibration of every unit before shipment.
- Compliance with electrical safety and EMC standards.
- Reliable sourcing of components to avoid supply chain interruptions.
- Efficient inventory management and production planning.

9.2 Selling Price Determination

The preliminary manufacturing cost of one prototype unit is estimated to be LKR 33,223. During volume production, the manufacturing cost is expected to decrease due to bulk purchasing, optimized production processes, and automated PCB assembly. However, the final selling price should also include expenses related to product assembly, testing, calibration, packaging, transportation, marketing, warranty services, distributor margins, and company profit.

Considering these factors and the advanced features provided by the Smart Power Quality Analyzer, the proposed selling price is:

$$\boxed{\text{Selling Price} = \text{USD } 300 \approx \text{LKR } 90,000} \quad (1)$$

This pricing is justified by the device's real-time power quality analysis, TFT graphical display, Wi-Fi connectivity, industrial-grade measurement capability, and long-term technical support.

9.3 Marketing, Sales and Beyond

The product will be promoted through technical demonstrations, engineering exhibitions, educational institutions, industrial partnerships, and online platforms. Customer support will include firmware updates, calibration services, warranty support, and technical assistance to ensure reliable long-term operation and customer satisfaction.

10 Project Budget with Bill of Quantities

Table 5: Preliminary bill of quantities

No.	Item	Quantity	Unit Cost (LKR)	Total Cost (LKR)
1	Electronic components used in one analyzer, including passive components	1 set	12,773	12,773
2	TFT display module	1	6,000	6,000
3	Connectors, terminals, and interconnecting cables	1 set	2,000	2,000
4	Product enclosure	1	3,000	3,000
5	Hi-Link isolated AC-DC power-supply module	1	1,450	1,450
6	PCB manufacturing	1 set	8,000	8,000
Estimated Total Project Cost				33,223

The estimated total project cost is calculated as

$$\begin{aligned}\text{Total Project Cost} &= 12,773 + 6,000 + 2,000 + 3,000 \\ &\quad + 1,450 + 8,000 \\ &= \boxed{\text{LKR } 33,223}.\end{aligned}\tag{2}$$

This estimate does not include shipping charges, customs duties, soldering expenses, testing equipment, or the cost of replacing damaged components during development.

11 Task Allocation

11.1 Individual Contributions

11.1.1 230045X - ANTHONY C.S.B.

Power supply and ESP32 circuit designing and component selection, PCB structure and module placement designing, Altium Designing, PCB component placement and PCB assembly, Soldering, ESP32 firmware development, LCD Display programming, PCB Testing, Hardware Assembly

11.1.2 230697V - WEERASINGHE J.A.H.R.

ADC and STM32 Circuit designing and component selection, Schematic design, Altium Designing, PCB component placement and PCB assembly, Soldering, STM32 firmware development, Software debugging, PCB Testing, Hardware Assembly

11.1.3 230100M - CHANDRAKUMARA H.A.D.C.

Component selection, Analog Signal Processing Circuit designing, Signal Acquisition circuit Design, Schematic design, Solidworks enclosure designing, PCB testing and PCB Assembly

11.1.4 230229P - HANSINDU M.M.A.D.

Power supply circuit Designing, Schematic designing, Altium designing, PCB Design Optimization, PCB testing and PCB Assembly, Component Selection

11.1.5 230477X - PERERA H.A.J.I.

Current and Voltage Sensing circuit Design, Enclosure, Web interface Design, Database integration for Data, Firmware modification and Debugging, Hardware and Software testing, LCD display code finalization, Documentation

References

- [1] IEEE, *IEEE Recommended Practice for Monitoring Electric Power Quality*, IEEE Standard 1159.
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- [4] Espressif Systems, *ESP32-S3 Series Datasheet*.
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